

Funkin' Machine (fnfmod) — build log summary

NeoForge 1.21.1 FNF rhythm mod. Compressed memo-to-self. Date range 2026-07-06..09.

Core / build

Gradle 9.1, ModDevGradle 2.0.141, neo 21.1.235, JDK21. jar: build/libs/fnfmod-1.0.0.jar. playerAnimator dep = dev.kosmx.player-anim:player-animation-lib-**forge**:2.0.4+1.21.1 (forge artifact IS the neoforge build on 1.21; -neoforge doesnt exist), maven.kosmx.dev, modid playeranimator, CLIENT-required.

Gotcha: PS5.1 Set-Content -Encoding utf8 writes BOM breaks javac (use UTF8Encoding(false)). Build via cmd /c gradlew to avoid PS NativeCommandError noise.

Song formats + scanning (SongLibrary)

3 chart formats: legacy FNF (wrapped {"song":{}}), Psych 1.0 psych_v1 (UNWRAPPED root, absolute lanes 0-3 player/4-7 opp vs pre-1.0 mustHitSection-relative — detect via format field), V-Slice (chart+metadata pair).

Songs from config/fnfmod/songs/<name>/ + external mod folders (Settings>Directories, external_folders.txt). Psych layout data/<song>/*.json + songs/<song>/*.ogg; V-Slice data/songs/<song>/. isModFolder = has data|songs|images.

V-Slice variations: <song>-chart[<var>].json (default/erect/pico/spookymod...) each own metadata + audio suffixed Inst-<var>.ogg. Vocals named after CHARACTER ID from playData.characters (Voices-<charId>[<var>].ogg), strip -playable (spooky-playable=Voices-spooky). NO bf/dad fallback for variations (was bf-grab bug); last-resort anyVoiceWithSuffix != opponent.

Mod merge: Polymod mods add variations to EXISTING base song ids. Scan MERGES across folders (no dup-skip). Deferred key assign: SongEntry.rawVars gathered, finalizeVSlice() assigns keys — default/unique diff = plain, name reused by many variations = ' (variation)' suffix (pico/spookymod easy/normal/hard). vsliceRealDiff maps display key back to real diff for parse; realDifficulty strips ' (x)' suffix for cache folders. transferFiles(diff)/allTransferFiles for MP streaming.

Multiplayer (SessionManager, server-authoritative)

Session keyed dim+BlockPos. First interactor = host picks song; guest 2nd click joins opp side. Chunked file streaming 400KB + SHA-1 manifest (integrated server never transfers; cache config/fnfmod/cache, pruned 30d unused). Start sync = RELATIVE delay not epoch. Note events relayed for partner strums/animations.

Stage: FunkinMachineBlock HORIZONTAL_FACING (points at placer). placeOnStage teleports 2 blocks front, +-1.5 along facing.getCounterClockWise (player=CCW=screen-right), yaw faces away (toward cam). Camera THIRD_PERSON_FRONT. Return pos+health+invuln restored on end (RETURN_POINTS/HEALTH_BEFORE/INVULN_BEFORE). Bot ArmorStand spawned solo (opp spot, angled toward player, invuln/noGrav/noBasePlate). Player invuln during song.

Gameplay (GameplayScreen)

Audio = raw OpenAL (STBVorbis decode, AL sources), inst=master clock, vocals resync >50ms drift. Judgements Sick/Good/Bad/Shit 45/90/135/166ms. Input judged at LIVE clock in key handler (liveSongPos) not cached per-frame (was 1-frame input lag).

Holds: release opens 0.5s grace (GameNote.releasedMs), re-press resumes AND hitAttempt hits overlapped note (allowGhostMiss!=resumed). Grace expire = full miss, trail vanishes. Totally-missed long note = gray+30% (NoteStyle.setMissed). Stacked-note protect: 1ms window (BOTH=40ms) same-lane co-credit.

Modes solo: Player/Opponent/Both. Opponent = strums stay FNF layout (not swapped), anims mirrored via playerAnimator MirrorModifier. Both = single centered strumline (middlescroll), all notes merged. Middlescroll option (opp notes to edges, 60% alpha). Beat zoom = FOV pinch every measure (ViewportEvent.ComputeFov). Camera X/Y pan via CameraMixin (Camera.setup TAIL), focus follows mustHitSection, per-section camEase (fnfmodCamEase field).

Skins / visuals (NoteStyle, SparrowAtlas)

Sparrow XML atlases: classic NOTE_assets.png+xml OR V-Slice notes+noteStrumline+noteSplashes+holdCover. NOTE_hold_assets.png (no xml) sliced INTERLEAVED cols [piece,end,piece,end]. Frame index = last 4 digits only (variant digit preserved: 'note splash red 10000'=variant1). rotated="true" frames un-rotated at draw.

Psych RGB template skins: red=base,green=highlight(white),blue=outline; per-lane remap into DynamicTexture. Applied notes/sustains/splashes/press+confirm receptors (NOT static). Note Colors picker (hue+SB+hex+toggle). Splash folder config/fnfmod/splashes (skin-own wins, disables selector).

RENDER GOTCHAS: (1) fill() batch flush DISABLES blend, blit uses leftover state — force RenderSystem.enableBlend+defaultBlendFunc before every sprite. (2) negative geometry scale flips winding = CULLED invisible — flip via negative UV width/height not pose scale (icons + hold trails). (3) Screen.render() runs blur shader — custom screens fill flat bg + iterate renderables, no super.render after content.

skin.json: noteScale/receptorScale/holdWidthScale/splashScale/holdCoverScale + noteAlpha/sustainAlpha/receptorAlpha/splashAlpha/holdCoverAlpha. Sprites sized by ART pixels (noteRefPx/strumRefPx) not logical box (press/confirm have glow padding). Hold cover placement from Funkin Strumline.hx: 104px strums, 0.7 scale, -12/+15.4 receptor-px offset.

Icons (IconLibrary) — PER MOD

Icon/char names clash across mods. Each song resolves opponent icon to a FILE from ITS OWN mod: SongEntry.modRoot (default variation mod) + resolveOpponentIcon reads modRoot char json healthIcon.id (V-Slice) / healthicon (Psych) then icon-<name>.png in modRoot images/icons|icons. SongInfo carries opponentIconPath (abs) — IconLibrary.getByPath/hasFile/drawFile keyed by path (MP falls back to named has()). IconLibrary auto-syncs via SongLibrary.rescanGeneration() counter (was stale after Settings add-folder).

HUD styles (Settings>Visuals)

default/abbreviated/numbers = XP-bar health + score text. vanilla = REAL hearts (SetHealthC2S server clamps [1,max] never lethal, restore after; forced->default in creative via effectiveHudStyle instabuild check) + replica xp(misses)+hunger(accuracy). fnf = hide ALL vanilla, Psych bar + icons (48px, player mirrored, OutCirc beat bop, bar colors from character.json healthbar_colors) + MC-font score. Downscroll (non-fnf): hotbar flush top (FnfClient translates HOTBAR layer -(guiH-22) Pre push/Post pop), hearts/hunger/xp cluster BELOW it, xp number always ABOVE bar. Song title+progress to bottom. FNF bar downscroll y=28 (mid, icons no clip). Bg dim removed.

UI screens

SongSelectScreen = scrollable GRID (BOX 84x92, responsive cols, pixel scroll+drag thumb). Box = opponent icon + name, LAYER 0x88000000 stacked darkens. Click opens SongDetailScreen (best score panel + Difficulty[dropdown list to left]/Play as/Play Solo/Play VS). Main keeps Editor/Settings/Reload/Close. ScoreStore = config/fnfmod/scores.json (songId|diff, Record{score,maxCombo,sick,good,bad,shit,missed,totalNotes}), saved on finish, shown detail+results. Chart editor /fnf editor: centered fixed playhead, grid scrolls under, wheel/W/S scrub (playback follows), vortex mode (keys 1-8 place at playhead), per-section MustHit/AltAnim/GF/ChangeBPM/beats/CamEase, P/O hitsound toggles, saves Psych format (cols 0-3 opp normalized). Settings sections: Note Colors / Controls / Adjust Delay / Visuals / Gameplay / Directories. Shift+click cycles backward. Rating Position drag screen. Sliders scroll speed 0.35-6 (Const/Mult toggle) + hitsound vol 0-100. Icon = searchable IconPickerScreen. Hint text above Back button.

Animations (CharacterAnimations, playerAnimator)

Sets = subfolders config/fnfmod/animations/<set> (loose=default). character.json: cameraOffset + per-anim {anim,cameraOffset}. Actions idle/idle2(danceLeft/Right alternate on beat)/left/down/up/right/miss. Play on real player entities both sides (set sent via ReadyC2S/StartSongS2C). Body aligned to head yaw during song (else torso lags anims). Looping idle not restarted (rest-pose flash); fade between anims (AbstractFadeModifier). Sing loops every 180ms while holding.

Test rigs (scratchpad): VSliceTest/LocalScanTest/MergeTest/IconTest call chart parser + SongLibrary standalone (gson+build/classes, no MC classpath since FMLPaths resolves lazily). Not yet real-playtested end-to-end by me.